

MAHAD JAFRI

Senior Data Scientist | Machine Learning | Experimentation & Production ML

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PROFESSIONAL SUMMARY

Senior Data Scientist with 5+ years building production ML systems and running controlled experiments that drive measurable business impact. Specialize in Bayesian modeling, time-series forecasting, reinforcement learning, and the production infrastructure to deploy them at scale. Proven impact: \$20M+ in client revenue lifted, 500K+ daily inferences served across telco and mobility.

CAREER HIGHLIGHTS

- **Revenue Impact:** Contributed to **\$20M+** in client revenue across telco and mobility, including **\$10M** lifted on Telefónica Spain through Bayesian pairing models validated via controlled A/B experiments.
- **Production ML:** Trained and deployed pairing-optimization models serving **~500K real-time inferences/day** on a 24/7 production platform with multi-account rollout.
- **Top Performer:** Recognized in the **top 10%** of Afiniti employees for **4 consecutive years** for technical leadership and revenue impact.

EXPERIENCE

Careem Technologies (Uber & e& subsidiary), Dubai (Remote)

Mar 2025 – Present

Senior Analytics & Automation Lead

- **Forecasting & Optimization:** Built a real-time supply-balancing system combining **LSTM and gradient-boosted forecasting** with reinforcement learning to generate targeted driver incentives, **cutting incentive campaign costs by 60% at WMAPE <10%**.
- **MLOps & HITL Workflows:** Built a Slack-integrated approval flow with checkpointing and state recovery to ship the supply-balancing system into ops workflows, **reducing manual execution time by 90%** while maintaining safety guarantees.
- **Experimentation:** Designed and executed an **A/B test on a new preferential care policy** in partnership with Care team stakeholders, communicating results to leadership that led to a **2% lift in on-time deliveries** and improved ETA adherence.
- **Conversational Analytics:** Built an **LLM-powered agent** that answers ad-hoc data questions with queries, charts, and written summaries, reducing executive data requests by **40%+** and freeing analyst capacity for higher-leverage modeling work.

Afiniti Software Solutions, USA (Remote)

Mar 2020 – Mar 2025

Senior Data Scientist

- **Bayesian Modeling:** Productionized MCMC (Stan) and decision-tree pairing models that **lifted client revenue by 5%**, validated through controlled A/B experiments.
- **Revenue Forecasting:** Built a 90-day forecasting pipeline using gradient-boosted ensembles with **85% accuracy** from 3-day behavioral signals to flag early-stage churn and automate LTV reporting.
- **Production ML at Scale:** Led a team of 4 to deliver **\$10M annual revenue lift** for Telefónica Spain; trained pairing-optimization models serving **~500K real-time inferences/day** on Afiniti's 24/7 production platform.
- **Modeling Infrastructure:** Enhanced training and validation modules for Afiniti's company-wide R modeling platform, **improving global model validation accuracy by 10%**.
- **Data Pipelines:** Optimized multi-stage PostgreSQL/MySQL attribution logic, **improving outcome-to-call match rate by 8%** and **reducing processing latency by 40%**.

TECHNICAL SKILLS

Machine Learning: Bayesian Inference (MCMC, Stan), Gradient Boosting (XGBoost, LightGBM), Random Forest, Deep Learning, LSTM & Sequence Models, Reinforcement Learning, Time-Series Forecasting, Causal Inference, NLP

Experimentation & Statistics: A/B Testing, Hypothesis Testing, Causal Inference, Power Analysis, Cohort Analysis, Statistical Modeling, Experiment Design

LLMs & GenAI: LLM Agents, LangGraph, LangChain, ReAct, RAG, Text-to-SQL, Prompt Engineering, Embeddings, Fine-tuning, OpenAI/Anthropic APIs

Engineering & Tools: Python, SQL (PostgreSQL, MySQL, Redshift), R, C++, PyTorch, scikit-learn, pandas, NumPy, FastAPI, Docker, Git, Tableau, Power BI, Superset

EDUCATION

Bachelor of Science, Electrical Engineering

Lahore University of Management Sciences, 2019